

## Graphical solutions



- 1 Consider the function  $y = \cos(x - 30)$ .
  - a Sketch the graph of the function for  $0 \leq x \leq 360$ .
  - b Sketch the line  $y = \frac{1}{2}$  on the same number plane.
  - c Hence, state all solutions to the equation  $\cos(x - 30) = \frac{1}{2}$  over the domain  $(0, 360]$ .
  
- 2 Consider the function  $y = \sin(x - 60)$ .
  - a Sketch the graph of the function for  $0 \leq x \leq 360$ .
  - b Sketch the line  $y = \frac{1}{2}$  on the same number plane.
  - c Hence, state all solutions to the equation  $\sin(x - 60) = \frac{1}{2}$  over the domain  $[0, 360)$ .
  
- 3 Consider the function  $y = 2 \sin 4x$ .
  - a Sketch the graph of the function for  $0 \leq x \leq 120$ .
  - b Sketch the line  $y = 1$  on the same number plane.
  - c Hence, state all solutions to the equation  $2 \sin 4x = 1$  over the domain  $[0, 90]$ . Give your answers in degrees.
  
- 4 Consider the function  $y = 2 \sin 2x$ .
  - a Sketch the graph of the function for  $0 \leq x \leq 180$ .
  - b State the other function you would add to the graph in order to solve the equation  $2 \sin 2x = 1$ .
  - c Sketch the graph of this function on the same number plane.
  - d Hence, state all solutions to the equation  $2 \sin 2x = 1$  over the domain  $[-180^\circ, 180^\circ]$ .
  
- 5 Consider the function  $y = 3 \cos 2x + 1$ .
  - a Sketch the graph of the function for  $0 \leq x \leq 180$ .
  - b State the other function you would add to the graph in order to solve the equation  $3 \cos 2x + 1 = \frac{5}{2}$ .
  - c Sketch the graph of this function on the same number plane.
  - d Hence, state all solutions to the equation  $3 \cos 2x + 1 = \frac{5}{2}$  over the domain  $[0, 180]$ .

6 Consider the function  $y = 2 \sin 3x - 3$ .



- a Sketch the graph of the function for  $0 \leq x \leq 60$ .
- b State the other function you would add to the graph in order to solve the equation  $2 \sin 3x - 3 = -2$ .
- c Sketch the graph of this function on the same number plane.
- d Hence, state all solutions to the equation  $2 \sin 3x - 3 = -2$  over the domain  $[0, 60]$ .

7 Consider the function  $y = -2 \cos 3x$ .

- a Sketch the graph of the function for  $0^\circ \leq x \leq 120^\circ$ .
- b State the other function you would add to the graph in order to solve the equation  $-2 \cos 3x = -1$ .
- c Sketch the graph of this function on the same number plane.
- d Hence, state all solutions to the equation  $-2 \cos 3x = -1$  over the domain  $[0^\circ, 120^\circ]$ .

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## Exact value equations

8 State whether the following equations have a solution:

a  $\cos \theta - 4 = 0$

b  $9 \tan \theta + 4 = 0$

9 State the number of solutions for  $\theta$  of the following equations in the domain  $0^\circ < \theta < 90^\circ$ .

a  $\cos \theta = -\frac{1}{\sqrt{2}}$

b  $\sin \theta = -\frac{\sqrt{3}}{2}$

c  $\tan \theta = -1$

10 Solve the following equations for  $0^\circ \leq \theta \leq 90^\circ$ :

a  $\sin \theta = \frac{1}{\sqrt{2}}$

b  $\tan \theta = \sqrt{3}$

c  $\cos \theta = \frac{1}{2}$

d  $\sin \theta = \frac{\sqrt{3}}{2}$

11 Solve the following equations for  $0^\circ \leq \theta \leq 360^\circ$ :



a  $\cos \theta = -\frac{1}{\sqrt{2}}$

c  $\cos \theta = 0$

e  $\sin \theta = 0$

g  $\cos \theta = -\frac{1}{\sqrt{2}}$

i  $\sin \theta = 1$

k  $\tan \theta = 0$

m  $4 \tan \theta + 2 = -2$

o  $2 \cos \theta + 4 = 3$

b  $\cos \theta = \frac{1}{2}$

d  $\sin \theta = \frac{1}{2}$

f  $\sin \theta = -\frac{1}{\sqrt{2}}$

h  $\sin \theta = -\frac{\sqrt{3}}{2}$

j  $\tan \theta = \sqrt{3}$

l  $\tan \theta = -\frac{1}{\sqrt{3}}$

n  $8 \cos \theta - 4 = 0$

p  $8 \sin \theta - 4\sqrt{2} = 0$

## Non-exact value equations

12 Solve the following equations for  $0^\circ \leq \theta \leq 90^\circ$ :

a  $\cos \theta = 0.7986$

b  $\sin \theta = 0.6428$

c  $\tan \theta = 0.7265$

d  $\sin \theta = 0.3584$

e  $\tan \theta = 2.2460$

13 Solve the following equations for  $0^\circ \leq \theta \leq 360^\circ$ :

a  $\cos \theta = 0.9063$

b  $\cos \theta = -0.7986$

c  $\sin \theta = -0.6428$

d  $\sin \theta = 0.9336$

e  $\tan \theta = 0.7002$

f  $\tan \theta = -0.7265$